DOMAĆA ZADAĆA BR 5.

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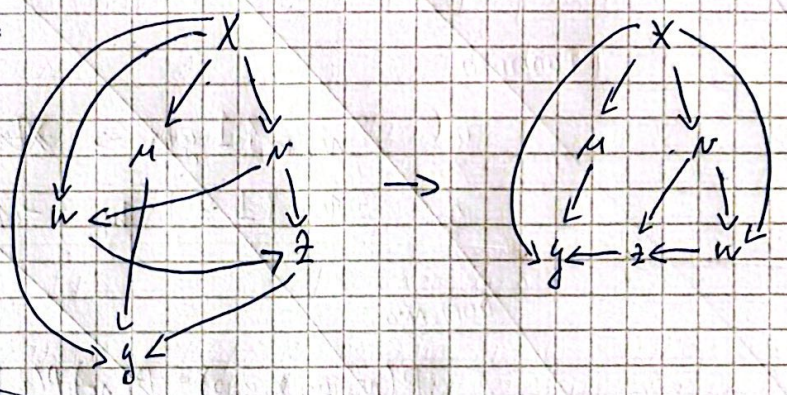
17.1.

$$P(u,v,w,x,y,z) = P(u|x)\,P(v|x)\,P(w|v,x)\,P(x)\,P(y|u,v,z)\,P(z|v,w)$$

CRTAM GRAF:



ZNAČI:

v i w → OPAŽENE

y → NE OPAŽENA

Bj: $\boxed{B}$: u → SUSJEDNO

STAZE od x do z:

1. $x \to v \to z$
   ↳ v bi bio bili OPAŽEN
2. $x \to w \to z$ → w. OPAŽEN
3. $x \to y \leftarrow z$ → y NEOPAŽEN ali u BLOKIRA
4. $x \to u \to y \leftarrow z$ → VEĆ OMES ZBOG y, ali ulazi u u

tvori: $x \to w \leftarrow v \to z$ je BLOKIRAN ZBOG w

Za u je $\boxed{\text{SUSJEDNO}}$

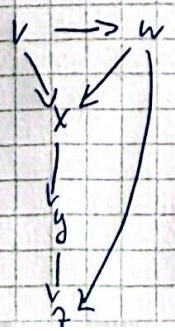
17.2.

- v, w, x, y, z

VRIJEDI: $\{v, w\} \perp y \mid x$

$\{v, x\} \perp z \mid \{w, y\}$

$$P(v,w,x,y,z) = P(v)\,P(w|v)\cdot P(x|v,w)\,P(y|v,w,x)\cdot P(z|x,w,v,y)$$
$$= P(v)\,P(w|v)\cdot P(x|v,w)\cdot P(y|x)\cdot P(z|y,w)$$



A $v \perp z \mid x$ ↳ NE GARANTIRA, POSTOJI PUT $v \to w \to z$ koji $v \to w$

B $x \perp z \mid w$ ↳ IMAMO $x \to y \to z$ koji je OTVOREN jer UVJETUJEMO NA w a ne y.

C $w \perp y \mid x$ → VRIJEDI jer je y DIJETE od x i NEMA PROLAZA PODUDELNA TAKO DA SVAKI PUT MORA UĆI U y PREKO x AKO DIJELI SE BLOKIRANI.

D $x \perp z \mid y$

Ispravan je $\boxed{C}$ $w \perp y \mid x$

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17.3

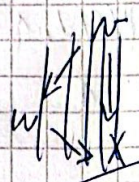
v, w, x, y, z

BINARNE

TERNARNE

POTPUNI DAG:

BROJ PARAMETARA	
$2=(2-1)\cdot 1$	$pa(v)=\emptyset$
3	$pa(w)=\{v\}$
12	$pa(x)=\{v,w\}$
3	$pa(y)=\{\cancel{v},\cancel{w},x\}$
4	$pa(z)=\{\cancel{v},\cancel{w},\cancel{x},y\}$



je $\{v,w\} \perp y \mid x$ — y NE OVISI O v, w, ako mu x

je $\{v,x\} \perp z \mid \{w,y\}$ — z NE OVISI O v, x ako su w, y

FAKTORIZACIJA:

$$P(v,w,x,y,z)=P(v)\,P(w\mid v)\cdot P(x\mid v,w)\cdot P(y\mid x)\cdot P(z\mid w,y)$$

BROJ PARAMETARA:

$$2+3+12+3+4=24$$

$\boxed{\text{Rj: } \boxed{B}\ 24}$

18.1

$$P(x,y,z,w)=P(z)\,P(w)\,P(y\mid z,w)\,P(x\mid y)$$

$N=10\,000$

$n(z=1)=0.2$
$n(w=1)=0.2$
$\Rightarrow P(w=0)=0.8$

$n(x=1\mid y=0)=0.2$
$\Rightarrow n(x=0\mid y=0)=0.8$
$n(x=1\mid y=1)=0.6$
$\Rightarrow P(x=0\mid y=1)=0.4$

IZ TABLICE

$n(y=1\mid z=1, w=0)=0.2$
$\Rightarrow P(y=0\mid 1,0)=0.8$
$n(y=1\mid z=1, w=1)=0.7$
$\Rightarrow P(y=0\mid 1,1)=0.3$

ODBACIVANJE za $z=1$, $z=0$

$$E[N_{w=0}-N_{w=1}]=N\left(P(w=0,x=0,z=1)-P(w=1,x=0,z=1)\right)$$

$$P(w,x=0,z=1)=P(z=1)\,P(w)\sum_{y\in\{0,1\}}P(y\mid z=1,w)\,P(x=0\mid y)$$

za $w=0$: $\sum_y = P(y=0\mid 1,0)\cdot 0.8+P(y=1\mid 1,0)\cdot 0.4 = 0.8\cdot 0.8+0.2\cdot 0.4 = 0.64+0.08=0.72$

$P(w=0,x=0,z=1)=0.2\cdot 0.8\cdot 0.72=0.1152$

za $w=1$: $\sum_y = P(y=0\mid 1,1)\,0.8+P(y=1\mid 1,1)\cdot 0.4=0.24+0.28=0.52$

$P(w=1,x=0,z=1)=0.2\cdot 0.2\cdot 0.52=0.0208$

KONAČNO:

$P(w=0,x=0,z=1)-P(w=1,x=0,z=1)=0.1152-0.0208=0.0944$

$E[N_{w=0}-N_{w=1}]=10000\cdot 0.0944=\boxed{944}$

$\boxed{\text{Rj: } \boxed{B}\ 944}$

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18.2 ZBOG STOHASTIČNOSTI MREŽE VRIJEDI:

$L \to S, N \to S, T$

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$$P(L,S,N) = \sum_T P(L)\cdot P(N)\cdot P(S|L,N)\, P(T|N)$$

$$= P(L)\, P(N)\cdot P(S|L,N)$$

jer je $\sum_T P(T|N) = 1$

$N = 7$, - MAP (LAPINCE)

$L =$ ~~AB~~ DOBRA, 1 je LOŠA $\qquad P(L=1) = \frac{4+1}{7+2} = \frac{5}{9}$

$N = 2$ je 2 crna $\qquad P(N=2) = \frac{3+1}{7+3} = \frac{4}{10} = \frac{2}{5}$

$P(S=1 \mid L=1, N=2)$ za $\alpha = \beta = 2$

$$P(S=1 \mid L=1, N=2) = \frac{1+1}{1+2} = \frac{2}{3}$$

KONAČNO: $P(L=1, S=1, N=2) = \frac{5}{9}\cdot\frac{2}{5}\cdot\frac{2}{3} = \frac{4}{27} \approx 0.148.$

Tj: $\boxed{B}$ 0.148

18.3 $\mu_{MAP} = ?$

$\mu = P(Y=0 \mid y=1, z=0)$

$N = 100$ uzoraka iz $P(w)P(x)P(y|w,x)P(z|x)$

PRIMIJENIMO SAMO ONE $\wedge\, y=1, \wedge\, z=0$

1.) $P(y=1, z=0)$

$$P(y=1, z=0) = \sum_w \sum_x P(w)P(x)P(y=1|w,x)P(z=0|x)$$

VRIJEDI: $P(w=1)=0.1 \Rightarrow P(w=0)=0.9$

$P(x=1)=0.2 \Rightarrow P(x=0)=0.8$

$P(z=0|x=0)=0.1 \quad P(z=0|x=1)=0.3$

$P(y=1|w,x)$: $(0,0)\to 0$, $(0,1)\to 0.4$, $(1,0)\to 0.2$, $(1,1)\to 0.7$

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POMNOŽI:

$w=0,\ x=0:\ 0.9\cdot 0.8\cdot 0.1\cdot 0.1=0$

$w=0,\ x=1:\ 0.9\cdot 0.2\cdot 0.4\cdot 0.3=0.0216$

$w=1,\ x=0:\ 0.1\cdot 0.8\cdot 0.2\cdot 0.1=0.0016$

$w=1,\ x=1:\ 0.1\cdot 0.2\cdot 0.7\cdot 0.3=0.0042$

$$P(y=1,\ z=0)=0.0276$$

$$P(x=0,\ y=1,\ z=0)=0.8\cdot 0.1(0.9\cdot 0+0.1\cdot 0.2)=0.0016$$

BROJ MEĐU NULA $\wedge\ x=0$

$$\hat{\mu}_{MAP}=\frac{k+1}{M+2}$$

$\hookrightarrow M$ - PRIHVAĆENI, $y=1, z=0$

za $N=100$

$$E[M]=N\cdot P(y=1,\ z=0)=100\cdot 0.0276=2.76$$

$$E[k]=N\cdot P(x=0,\ y=1,\ z=0)=100\cdot 0.0016=0.16$$

FINALNO:

$$E[\hat{\mu}_{MAP}]\approx\frac{0.16+1}{2.76+2}=0.2447$$

Rješenje:

$\boxed{B}$ 0.2447